

HR Analytics - Predict Employee Attrition

INTRODUCTION

Employee attrition is a major challenge faced by organizations, leading to increased hiring costs and loss of experienced employees. Predicting which employees are likely to leave helps companies take proactive measures to improve retention. This project focuses on analyzing employee data and building a machine learning model to predict attrition.

ABSTRACT

This project aims to develop a machine learning-based system to predict employee attrition using historical HR data. The dataset contains various employee attributes such as age, salary, job role, overtime status, and years at the company. A Random Forest classifier is used to predict whether an employee will stay or leave. Additionally, SHAP (SHapley Additive exPlanations) is used to interpret model decisions and identify key factors influencing attrition. The system helps HR departments understand employee behavior and improve retention strategies.

TOOLS USED

- Python
 - Pandas & NumPy (Data Processing)
 - Matplotlib & Seaborn (Data Visualization)
 - Scikit-learn (Machine Learning)
 - SHAP (Model Explainability)
 - Power BI (Dashboard Visualization)
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STEPS INVOLVED IN BUILDING THE PROJECT

1. Data Collection

The HR dataset containing employee details and attrition status was loaded for analysis.

2. Data Preprocessing

Missing values were checked, and categorical variables were converted into numerical format using Label Encoding.

3. Exploratory Data Analysis (EDA)

Data visualization was performed to understand patterns such as attrition distribution, salary impact, and overtime influence.

4. Model Building

A Random Forest Classifier was trained using an 80-20 train-test split to predict employee attrition.

5. Model Evaluation

The model achieved approximately 85% accuracy. Performance was evaluated using classification metrics and confusion matrix.

6. Explainable AI (SHAP Analysis)

SHAP was used to interpret predictions and identify important factors such as overtime, monthly income, age, and years at the company.

7. Visualization

Power BI was used to create interactive dashboards for better business understanding of attrition trends.

CONCLUSION

This project successfully demonstrates a machine learning approach to predict employee attrition. The model helps identify employees at risk of leaving and highlights key factors influencing attrition such as overtime, income, and experience. These insights can assist HR departments in making data-driven decisions to improve employee retention and organizational efficiency.

Additionally, the use of SHAP explainability provides transparency in model predictions by showing why a particular employee is likely to stay or leave. This makes the model more reliable and interpretable for real-world HR applications. The project also shows how machine learning can be effectively applied to solve business problems in human resource management.